

Case Study- 1

CitePrism: LLM-Driven Audit of Self-Citations and Reference Relevance

Under the guidance of:
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Github : <https://github.com/SRH-Heidelberg-University-ADSA/CitePrism>

Introduction

CitePrism is a modern system for auditing academic manuscripts on citation relevance, with a focus on evaluating self-citations. By leveraging Large Language Models (LLMs) and embedding techniques, CitePrism ensures clarity in scholarly practices by detecting off-topic citations, ranking references by relevance, and generating audit-ready reports.



CITEPRISM
WHERE CITATIONS
MEET INTELLIGENCE

Problem Statement

Checking every reference in a research paper is too slow for humans to do alone. Because of this, some authors 'cheat' by citing themselves or their friends just to boost their scores, even if the work isn't relevant. This makes academic research less trustworthy.



User story



As a research checker,

I want an AI dashboard that quickly spots useless citations,

So that I can make sure every reference is actually needed without doing all the work by hand.

Business Questions

1. How does CitePrism improve audit efficiency?

- The Problem: Manual citation review currently takes 3–4 hours per manuscript.
- The Result: CitePrism reduces this time to 30–45 minutes, allowing a single analyst to process 4x–5x more papers every day.

2. How do we prevent "False Positives" (wrongly flagging good citations)?

- The Solution: The system doesn't just look for "self-citations"; it combines relevance scoring with contextual analysis.
- Human-in-the-Loop: Only citations that fall below a specific user-defined threshold (τ) trigger a flag, ensuring high-quality research isn't penalized by mistake

3. Can the AI's decisions be challenged by a human?

- Transparency: Every score includes quoted evidence and a clear LLM-generated rationale.
- Control: Analysts can manually adjust the threshold, review the AI's logic, and override any automated decision based on their own expertise.

Core Objectives



Detection

Identify self-citations via author, team, or venue overlap with precise attribution



Assessment

Rank every reference on a 0–100 relevance scale with evidence-based scoring



Interactivity

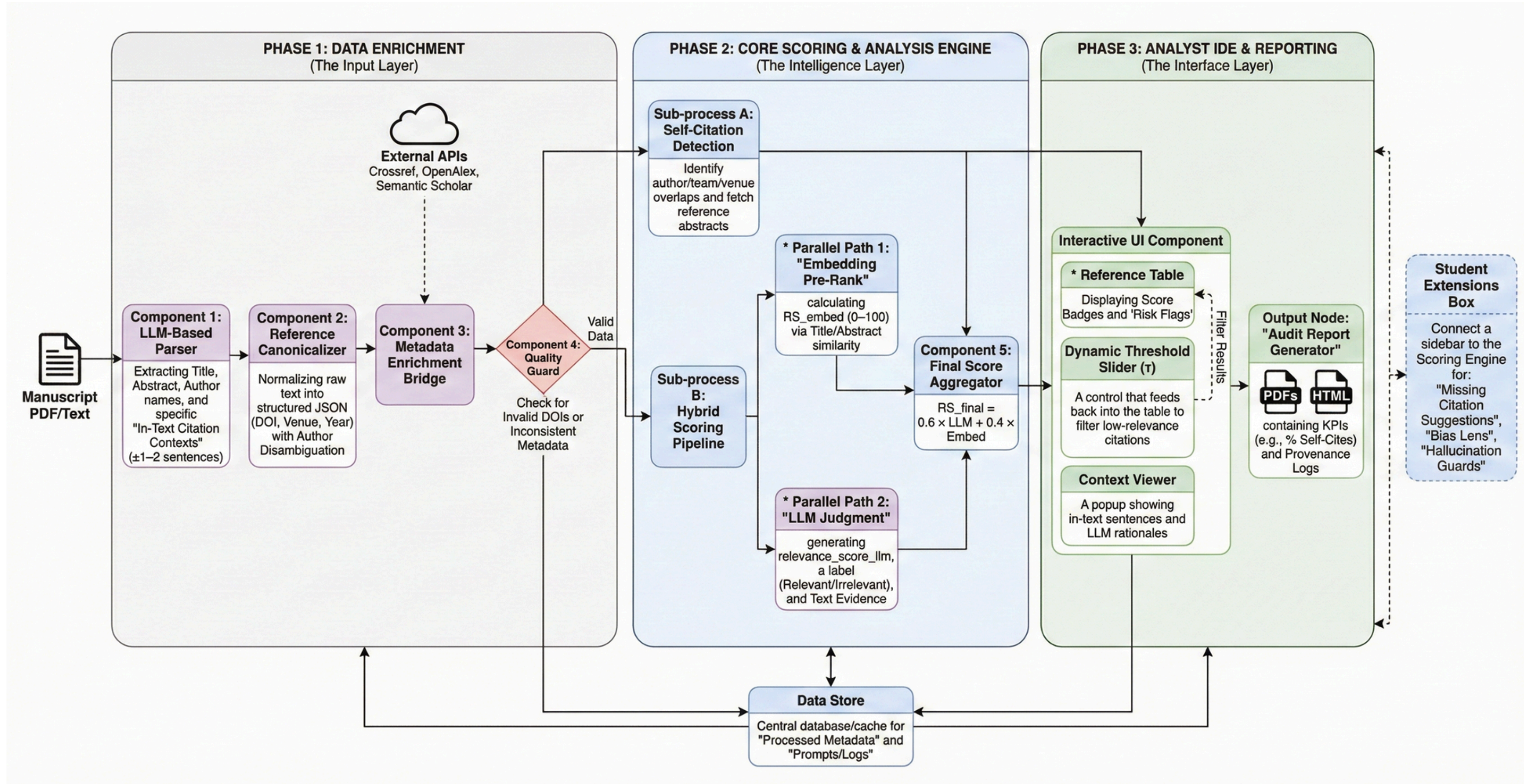
Empower analysts with UI slider (τ) to dynamically flag low-relevance citations



Auditability

Generate evidence-based reports with clear KPIs and provenance for reproducibility

Architecture



Live demonstration

System Status

Total Historical Audits

6

↑ Stored in Local DB

Core AI Engine

gemini-2.5-flash

↑ High-Speed Extraction

Scoring Paradigm

Hybrid

↑ LLM (60%) + Vectors (40%)



CitePrism

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Start Data Mining

Future scope

- Multi-Model Consensus Scoring: Implement an ensemble scoring system where multiple LLMs (e.g., Gemini, Llama, and Claude) cross-validate citation relevance to eliminate individual model bias and improve overall audit reliability.
- Springer and ScienceDirect API Integration: Expand the metadata enrichment pipeline by integrating official Springer and ScienceDirect API keys to retrieve full-text access and high-fidelity abstracts, directly addressing the current "Missing Abstract" bottleneck.
- Autonomous AI Audit Agents: Develop specialized AI agents capable of autonomously searching the web to resolve metadata hallucinations and "Hallucination Guard" mismatches without requiring manual human intervention.
- Explainable AI (XAI) Dashboards: Integrate SHAP or LIME-based visualization layers to provide transparent, feature-level explanations for why specific references were flagged, transforming "Black Box" scores into actionable editorial insights.

Thank you!

Any Questions ?